

WINKLER COUNTY AP, TX, USA

WMO: 722656

Lat: 31.780N

Lon: 103.202W

Elev: 856

StdP: 91.46

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.2	-18.1	0.8	9.5	-16.1	1.0	9.9	13.9	15.6	12.0	15.9	2.5	350	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	40.8	19.1	39.1	19.3	37.8	19.4	22.7	31.8	22.2	31.4	21.8	31.0	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	17.3	24.6	20.1	16.4	24.5	19.4	15.7	24.6	71.3	31.8	69.3	31.3	67.6	30.8	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.6	9.9	8.6	DB	-10.2	43.3	2.3	1.5	-11.9	44.4	-13.2	45.3	-14.5	46.1	-16.2	47.2
(5)				WB	-11.5	24.0	2.2	0.8	-13.0	24.5	-14.3	25.0	-15.5	25.4	-17.1	26.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	7.8	10.4	14.7	19.5	24.5	29.0	29.5	29.0	25.1	19.3	12.6	7.8
	DBStd	8.97	4.66	5.00	4.88	4.45	4.09	2.82	2.33	2.37	3.40	4.53	4.69	4.50
	HDD10.0	292	99	51	16	1	0	0	0	0	0	2	28	96
	HDD18.3	1276	328	222	129	39	6	0	0	0	2	43	178	327
	CDD10.0	3630	29	63	162	286	448	571	605	590	454	290	104	28
	CDD18.3	1571	0	1	17	74	196	321	347	332	206	72	5	0
	CDH23.3	21079	33	114	456	1229	2662	4331	4620	4244	2301	894	165	31
	CDH26.7	11869	3	25	159	589	1530	2667	2779	2504	1199	374	38	2
Wind	WSAvg	4.2	3.7	4.1	4.5	4.9	4.7	4.8	4.3	4.0	3.9	3.9	3.6	3.6
Precipitation	PrecAvg	292	10	12	12	16	24	40	45	34	50	31	11	10
	PrecMax	575	37	38	76	121	220	123	149	78	230	143	60	65
	PrecMin	116	0	0	0	0	0	0	0	2	0	0	0	0
	PrecStd	102	11	11	17	24	39	33	43	22	58	38	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	29.7	32.9	36.5	41.5	43.1	41.8	41.3	39.0	35.2	30.5	26.4
		MCWB	11.1	12.1	13.6	15.3	17.5	18.3	20.2	20.6	19.4	17.8	14.5	11.3
	2%	DB	23.8	26.8	30.7	34.4	38.8	41.1	39.7	39.1	36.6	32.8	27.4	23.6
		MCWB	10.3	11.4	13.1	14.4	16.7	18.7	20.1	20.3	19.2	17.2	13.2	10.8
	5%	DB	21.0	24.1	28.3	32.5	36.8	39.3	38.2	37.6	34.9	31.0	25.1	20.3
		MCWB	9.2	10.5	12.4	14.0	16.5	18.8	20.2	20.3	19.3	16.6	12.6	9.5
	10%	DB	17.8	21.2	26.0	30.4	34.5	37.4	37.0	36.3	33.0	28.6	22.6	17.5
		MCWB	8.0	9.5	11.8	13.6	16.0	19.2	20.2	20.3	19.2	16.1	12.0	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	14.8	17.8	19.6	21.9	23.5	23.6	23.2	22.5	21.7	17.9	14.1
		MCDB	20.0	21.1	24.9	27.5	30.5	32.2	31.8	32.8	30.5	28.8	22.7	19.2
	2%	WB	11.6	13.1	16.0	18.3	20.8	22.5	22.6	22.5	21.8	20.5	15.7	11.9
		MCDB	19.9	21.4	22.8	25.6	29.0	31.9	31.4	32.1	29.7	27.2	21.7	19.9
	5%	WB	10.1	11.8	14.7	17.1	20.1	21.9	22.1	22.0	21.3	19.5	14.6	10.5
		MCDB	19.3	21.2	22.5	25.2	28.3	31.2	31.4	31.6	29.1	25.6	20.8	19.2
	10%	WB	8.6	10.5	13.2	15.8	19.2	21.3	21.6	21.6	20.8	18.4	13.2	9.0
		MCDB	17.4	20.3	22.6	24.3	27.9	30.8	31.3	31.2	28.5	24.3	20.9	17.0
Mean Daily Temperature Range		MDBR	16.8	17.2	17.5	17.9	16.7	15.5	14.0	13.8	13.9	15.6	16.5	16.3
		MCDBR	22.5	22.6	21.9	20.7	20.1	18.2	16.0	15.5	16.6	18.8	20.8	21.8
		MCWBR	11.5	10.7	8.8	8.0	6.7	4.8	4.0	3.9	4.7	6.6	9.4	11.3
	5% WB	MCDDBR	19.4	19.0	17.5	16.8	15.9	14.9	13.6	13.5	12.9	13.0	15.6	18.8
		MCWBR	10.2	9.5	8.1	7.5	6.5	4.6	3.7	3.6	4.0	5.2	7.5	10.2
Clear-Sky Solar Irradiance	taub	0.261	0.269	0.287	0.324	0.342	0.375	0.397	0.371	0.354	0.305	0.282	0.266	
	taud	2.585	2.565	2.520	2.398	2.410	2.371	2.350	2.405	2.425	2.557	2.552	2.568	
	Ebn at Noon	983	1002	999	965	943	907	885	908	914	947	947	958	
	Edh at Noon	79	89	101	119	119	123	125	117	110	89	81	76	
All-Sky Solar Radiation	RadAvg	3.53	4.41	5.68	6.92	7.39	7.63	7.23	6.71	5.67	4.68	3.71	3.17	
	RadStd	0.33	0.42	0.39	0.41	0.45	0.35	0.35	0.30	0.37	0.45	0.36	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page